

UNITS AND DIMENSIONS

Learning objective: After going through this chapter, students will be able to;

- *understand physical quantities, fundamental and derived;*
- *describe different systems of units;*
- *define dimensions and formulate dimensional formulae;*
- *write dimensionalequations and apply these to verify various formulations.*

1.1 DEFINITION OF PHYSICS AND PHYSICAL QUANTITIES

Physics: Physics is the branch of science, which deals with the study of nature and properties of matter and energy. The subject matter of physics includes heat, light, sound, electricity, magnetism and the structure of atoms.

For designing a law of physics, a scientific method is followed which includes the verifications with experiments. The physics, attempts are made to measure the quantities with the best accuracy. Thus, Physics can also be defined as **science of measurement**.

Applied Physics is the application of the Physics to help human beings and solving their problem, it is usually considered as a bridge or a connection between Physics & Engineering.

Physical Quantities: All quantities in terms of which laws of physics can be expressed and which can be measured are called Physical Quantities.

For example; Distance, Speed, Mass, Force etc.

1.2 UNITS: FUNDAMENTAL AND DERIVED UNITS

Measurement: In our daily life, we need to express and compare the magnitude of different quantities; this can be done only by measuring them.

Measurement is the comparison of an unknown physical quantity with a known fixed physical quantity.

Unit: The known fixed physical quantity is called unit.

OR

The quantity used as standard for measurement is called unit.

For example, when we say that length of the class room is 8 metre. We compare the length of class room with standard quantity of length called metre.

Length of class room = 8 metre

$$Q = nu$$

Physical Quantity = Numerical value $\times$ unit

Q = Physical Quantity

n = Numerical value

u = Standard unit

e.g. Mass of stool = 15 kg

Mass = Physical quantity

15 = Numerical value

Kg = Standard unit

Means mass of stool is 15 times of known quantity i.e. Kg.

Characteristics of Standard Unit: A unit selected for measuring a physical quantity should have the following properties

- (i) It should be well defined i.e. its concept should be clear.
- (ii) It should not change with change in physical conditions like temperature, pressure, stress etc..
- (iii) It should be suitable in size; neither too large nor too small.
- (iv) It should not change with place or time.
- (v) It should be reproducible.
- (vi) It should be internationally accepted.

Classification of Units: Units can be classified into two categories.

- **Fundamental**
- **Derived**

Fundamental Quantity: *The quantity which is independent of other physical quantities.* In mechanics, mass, length and time are called fundamental quantities. Units of these fundamental physical quantities are called **Fundamental units**.

e.g. Fundamental Physical Quantity	Fundamental unit
Mass	Kg, Gram, Pound
Length	Metre, Centimetre, Foot
Time	Second

Derived Quantity: *The quantity which is derived from the fundamental quantities e.g. area is a derived quantity.*

$$\begin{aligned}\text{Area} &= \text{Length} \times \text{Breadth} \\ &= \text{Length} \times \text{Length} \\ &= (\text{Length})^2 \\ \text{Speed} &= \text{Distance} / \text{Time} \\ &= \text{Length} / \text{Time}\end{aligned}$$

The units for derived quantities are called **Derived Units**.

1.3 SYSTEMS OF UNITS: CGS, FPS, MKS, SI

For measurement of physical quantities, the following systems are commonly used:-

- (i) **C.G.S system:** In this system, the unit of length is centimetre, the unit of mass is gram and the unit of time is second.
- (ii) **F.P.S system:** In this system, the unit of length is foot, the unit of mass is pound and the unit of time is second.
- (iii) **M.K.S:** In this system, the unit of length is metre, unit of mass is kg and the unit of time is second.
- (iv) **S.I System:** This system is an improved and extended version of M.K.S system of units. It is called international system of unit.

With the development of science & technology, the three fundamental quantities like mass, length & time were not sufficient as many other quantities like electric current, heat etc. were introduced.

Therefore, more fundamental units in addition to the units of mass, length and time are required.

Thus, MKS system was modified with addition of four other fundamental quantities and two supplementary quantities.

Table of Fundamental Units

Sr. No.	Name of Physical Quantity	Unit	Symbol
1	Length	Metre	m
2	Mass	Kilogram	Kg
3	Time	Second	s
4	Temperature	Kelvin	K
5	Electric Current	Ampere	A
6	Luminous Intensity	Candela	Cd
7	Quantity of Matter	Mole	mol

Table of Supplementary unit

Sr. No	Name of Physical Quantity	Unit	Symbol
1	Plane angle	Radian	rad
2	Solid angle	Steradian	sr

Advantage of S.I. system:

- (i) It is coherent system of unit i.e. the derived units of a physical quantities are easily obtained by multiplication or division of fundamental units.
- (ii) It is a rational system of units i.e. it uses only one unit for one physical quantity. e.g. It uses Joule (J) as unit for all types of energies (heat, light, mechanical).
- (iii) It is metric system of units i.e. it's multiples & submultiples can be expressed in power of 10.

Definition of Basic and Supplementary Unit of S.I.

1. **Metre (m)**: The metre is the length of the path travelled by light in vacuum during a time interval of $1/299\,792\,458$ of a second.
2. **Kilogram (Kg)** : The kilogram is the mass of the platinum-iridium prototype which was approved by the Conférence Générale des Poids et Mesures, held in Paris in 1889, and kept by the Bureau International des Poids et Mesures.
3. **Second (s)**: The second is the duration of 9192631770 periods of the radiation corresponding to the transition between two hyperfine levels of the ground state of Cesium-133 atom.
4. **Ampere (A)** : The ampere is the intensity of a constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre apart in vacuum, would produce between these conductors a force equal to 2×10^{-7} Newton per metre of length.
5. **Kelvin (K)**: Kelvin is the fraction $1/273.16$ of the thermodynamic temperature of the triple point of water.
6. **Candela (Cd)**: The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of $1/683$ watt per steradian.
7. **Mole (mol)**: The mole is the amount of substance of a system which contains as many elementary entities as there are atoms in 0.012 kilogram of Carbon-12.

Supplementary units:

1. **Radian (rad)**: It is supplementary unit of plane angle. It is the plane angle subtended at the centre of a circle by an arc of the circle equal to the radius of the circle. It is denoted by θ .
$$\theta = l / r; \text{ } l \text{ is length of the arc and } r \text{ is radius of the circle}$$
2. **Steradian (Sr)**: It is supplementary unit of solid angle. It is the angle subtended at the centre of a sphere by a surface area of the sphere having magnitude equal to the square of the radius of the sphere. It is denoted by Ω .

$$\Omega = \Delta s / r^2$$

SOME IMPORTANT ABBREVIATIONS

Symbol	Prefix	Multiplier	Symbol	Prefix	Multiplier
D	Deci	10^{-1}	da	deca	10^1
c	centi	10^{-2}	h	hecto	10^2
m	milli	10^{-3}	k	kilo	10^3
μ	micro	10^{-6}	M	mega	10^6
n	nano	10^{-9}	G	giga	10^9

P	Pico	10^{-12}	T	tera	10^{12}
f	femto	10^{-15}	P	Pecta	10^{15}
a	atto	10^{-18}	E	exa	10^{18}

Some Important Units of Length:

- (i) 1 micron = 10^{-6} m = 10^{-4} cm
- (ii) 1 angstrom = $1\text{\AA} = 10^{-10}$ m = 10^{-8} cm
- (iii) 1 fermi = 1 fm = 10^{-15} m
- (iv) 1 Light year = 1 ly = 9.46×10^{15} m
- (v) 1 Parsec = 1 pc = 3.26 light year

Some conversion factor of mass:

- 1 Kilogram = 2.2046 pound
- 1 Pound = 453.6 gram
- 1 kilogram = 1000 gram
- 1 milligram = $1/1000$ gram = 10^{-3} gram
- 1 centigram = $1/100$ gram = 10^{-2} gram
- 1 decigram = $1/10$ gram
- 1 quintal = 100 kg
- 1 metric ton = 1000 kilogram

1.4 DEFINITION OF DIMENSIONS

Dimensions: The powers, to which the fundamental units of mass, length and time written as M, L and T are raised, which include their nature and not their magnitude.

For example

$$\text{Area} = \text{Length} \times \text{Breadth}$$

$$= [L^1] \times [L^1] = [L^2] = [M^0 L^2 T^0]$$

Power (0,2,0) of fundamental units are called dimensions of area in mass, length and time respectively.

e.g. Density = mass/volume

$$= [M]/[L^3]$$

$$= [M^1 L^{-3} T^0]$$

1.5 DIMENSIONAL FORMULAE AND SI UNITS OF PHYSICAL QUANTITIES

Dimensional Formula: An expression along with power of mass, length & time which indicates how physical quantity depends upon fundamental physical quantity.

e.g. Speed = Distance/Time

$$= [L^1]/[T^1] = [M^0L^1T^{-1}]$$

It tells us that speed depends upon L & T. It does not depend upon M.

Dimensional Equation: An equation obtained by equating the physical quantity with its dimensional formula is called dimensional equation.

The dimensional equation of area, density & velocity are given as under-

$$\text{Area} = [M^0L^2T^0]$$

$$\text{Density} = [M^1L^{-3}T^0]$$

$$\text{Velocity} = [M^0L^1T^{-1}]$$

Dimensional formula SI & CGS unit of Physical Quantities

Sr. No.	Physical Quantity	Formula	Dimensions	Name of S.I unit
1	Force	Mass $\times$ acceleration	$[M^1L^1T^{-2}]$	Newton (N)
2	Work	Force $\times$ distance	$[M^1L^2T^{-2}]$	Joule (J)
3	Power	Work / time	$[M^1L^2T^{-3}]$	Watt (W)
4	Energy (all form)	Stored work	$[M^1L^2T^{-2}]$	Joule (J)
5	Pressure, Stress	Force/area	$[M^1L^{-1}T^{-2}]$	Nm ⁻²
6	Momentum	Mass $\times$ velocity	$[M^1L^1T^{-1}]$	Kgms ⁻¹
7	Moment of force	Force $\times$ distance	$[M^1L^2T^{-2}]$	Nm
8	Impulse	Force $\times$ time	$[M^1L^1T^{-1}]$	Ns
9	Strain	Change in dimension / Original dimension	$[M^0L^0T^0]$	No unit
10	Modulus of elasticity	Stress / Strain	$[M^1L^{-1}T^{-2}]$	Nm ⁻²
11	Surface energy	Energy / Area	$[M^1L^0T^{-2}]$	Joule/m ²
12	Surface Tension	Force / Length	$[M^1L^0T^{-2}]$	N/m
13	Co-efficient of viscosity	Force $\times$ Distance / Area $\times$ Velocity	$[M^1L^{-1}T^{-1}]$	N/m ²
14	Moment of inertia	Mass $\times$ (radius of gyration) ²	$[M^1L^2T^0]$	Kg-m ²
15	Angular Velocity	Angle / time	$[M^0L^0T^{-1}]$	Rad.per sec
16	Frequency	1/Time period	$[M^0L^0T^{-1}]$	Hertz
17	Area	Length $\times$ Breadth	$[M^0L^2T^0]$	Metre ²
18	Volume	Length $\times$ breadth $\times$ height	$[M^0L^3T^0]$	Metre ³

19	Density	Mass/ volume	$[M^1L^{-3}T^0]$	Kg/m^3
20	Speed or velocity	Distance/ time	$[M^0L^1T^{-1}]$	m/s
21	Acceleration	Velocity/time	$[M^0L^1T^{-2}]$	m/s^2
22	Pressure	Force/area	$[M^1L^{-1}T^{-2}]$	N/m^2

Classification of Physical Quantity: Physical quantity has been classified into following four categories on the basis of dimensional analysis.

1. **Dimensional Constant:** These are the physical quantities which possess dimensions and have constant (fixed) value.

e.g. Planck's constant, gas constant, universal gravitational constant etc.

2. **Dimensional Variable:** These are the physical quantities which possess dimensions and do not have fixed value.

e.g. velocity, acceleration, force etc.

3. **Dimensionless Constant:** These are the physical quantities which do not possess dimensions but have constant (fixed) value.

e.g. e, π , numbers like 1, 2, 3, 4, 5 etc.

4. **Dimensionless Variable:** These are the physical quantities which do not possess dimensions and have variable value.

e.g. angle, strain, specific gravity etc.

Example.1 Derive the dimensional formula of following Quantity & write down their dimensions.

(i) Density

(ii) Power

(iii) Co-efficient of viscosity

(iv) Angle

Sol. (i) Density = mass/volume

$$=[M]/[L^3] = [M^1L^{-3}T^0]$$

(ii) Power = Work/Time

$$=\text{Force} \times \text{Distance}/\text{Time}$$

$$=[M^1L^1T^{-2}] \times [L]/[T]$$

$$=[M^1L^2T^{-3}]$$

(iii) Co-efficient of viscosity = $\frac{\text{Force} \times \text{Distance}}{\text{Area} \times \text{Velocity}}$

$$\frac{\text{Mass} \times \text{Acceleration} \times \text{Distance} \times \text{time}}{\text{length} \times \text{length} \times \text{Displacement}}$$

$$=[M] \times [LT^{-2}] \times [L] [T]/[L^2] \times [L]$$

$$=[M^1L^{-1}T^{-1}]$$

(iv) Angle = arc (length)/radius (length)

$$=[L]/[L]$$

$$=[M^0L^0T^0] = \text{no dimension}$$

Example.2 Explain which of the following pair of physical quantities have the same dimension:

(i) Work & Power (ii) Stress & Pressure (iii) Momentum & Impulse

Sol. (i) Dimension of work = force x distance = $[M^1L^2T^{-2}]$

Dimension of power = work / time = $[M^1L^2T^{-3}]$

Work and Power have not the same dimensions.

(ii) Dimension of stress = force / area = $[M^1L^{-1}T^{-2}]/[L^2] = [M^1L^{-1}T^{-2}]$

Dimension of pressure = force / area = $[M^1L^{-1}T^{-2}]/[L^2] = [M^1L^{-1}T^{-2}]$

Stress and pressure have the same dimension.

(iii) Dimension of momentum = mass x velocity = $[M^1L^1T^{-1}]$

Dimension of impulse = force x time = $[M^1L^1T^{-1}]$

Momentum and impulse have the same dimension.

1.6 PRINCIPLE OF HOMOGENEITY OF DIMENSIONS

It states that ***the dimensions of all the terms on both sides of an equation must be the same***. According to the principle of homogeneity, the comparison, addition & subtraction of all physical quantities is possible only if they are of the same nature i.e., they have the same dimensions.

If the power of M, L and T on two sides of the given equation are same, then the physical equation is correct otherwise not. Therefore, this principle is very helpful to check the correctness of a physical equation.

Example: A physical relation must be dimensionally homogeneous, i.e., all the terms on both sides of the equation must have the same dimensions.

In the equation, $S = ut + \frac{1}{2}at^2$

The length (S) has been equated to velocity (u) & time (t), which at first seems to be meaningless, But if this equation is dimensionally homogeneous, i.e., the dimensions of all the terms on both sides are the same, then it has physical meaning.

Now, dimensions of various quantities in the equation are:

Distance, $S = [L^1]$

Velocity, $u = [L^1T^{-1}]$

Time, $t = [T^1]$

Acceleration, $a = [L^1T^{-2}]$

$\frac{1}{2}$ is a constant and has no dimensions.

Thus, the dimensions of the term on L.H.S. is $S=[L^1]$ and

Dimensions of terms on R.H.S.

$$ut + \frac{1}{2}at^2 = [L^1T^{-1}] [T^1] + [L^1T^{-2}] [T^2] = [L^1] + [L^1]$$

Here, the dimensions of all the terms on both sides of the equation are the same. Therefore, the equation is dimensionally homogeneous.

1.7 DIMENSIONAL EQUATIONS, APPLICATIONS OF DIMENSIONAL EQUATIONS;

Dimensional Analysis: A careful examination of the dimensions of various quantities involved in a physical relation is called dimensional analysis. The analysis of the dimensions of a physical quantity is of great help to us in a number of ways as discussed under the uses of dimensional equations.

Uses of dimensional equation: The principle of homogeneity & dimensional analysis has put to the following uses:

- (i) Checking the correctness of physical equation.
 - (ii) To convert a physical quantity from one system of units into another.
 - (iii) To derive relation among various physical quantities.
1. **To check the correctness of Physical relations:** According to principle of Homogeneity of dimensions a physical relation or equation is correct, if the dimensions of all the terms on both sides of the equation are the same. If the dimensions of even one term differs from those of others, the equation is not correct.

Example 3. Check the correctness of the following formulae by dimensional analysis.

(i) $F = mv^2/r$ (ii) $t = 2\pi\sqrt{l/g}$

Where all the letters have their usual meanings.

Sol. $F = mv^2/r$

Dimensions of the term on L.H.S

Force, $F = [M^1L^1T^{-2}]$

Dimensions of the term on R.H.S

$$\begin{aligned}mv^2/r &= [M^1][L^1T^{-1}]^2 / [L] \\&= [M^1L^2T^{-2}] / [L] \\&= [M^1L^1T^{-2}]\end{aligned}$$

The dimensions of the term on the L.H.S are equal to the dimensions of the term on R.H.S. Therefore, the relation is correct.

$$(ii) \mathbf{t = 2\pi\sqrt{l/g}}$$

Here, Dimensions of L.H.S, $t = [T^1] = [M^0 L^0 T^1]$

Dimensions of the terms on R.H.S

Dimensions of (length) = $[L^1]$

Dimensions of g (acc due to gravity) = $[L^1 T^{-2}]$

2π being constant have no dimensions.

Hence, the dimensions of terms $2\pi\sqrt{l/g}$ on R.H.S

$$= (L^1 / L^1 T^{-2})^{1/2} = [T^1] = [M^0 L^0 T^1]$$

Thus, the dimensions of the terms on both sides of the relation are the same i.e., $[M^0 L^0 T^1]$. Therefore, the relation is correct.

Example 4. Check the correctness of the following equation on the basis of dimensional analysis, $V = \sqrt{\frac{E}{d}}$. Here V is the velocity of sound, E is the elasticity and d is the density of the medium.

Sol. Here, Dimensions of the term on L.H.S

$$V = [M^0 L^1 T^{-1}]$$

Dimensions of elasticity, $E = [M^1 L^{-1} T^{-2}]$

& Dimensions of density, $d = [M^1 L^{-3} T^0]$

Therefore, Dimensions of the terms on R.H.S

$$\sqrt{\frac{E}{d}} = [M^1 L^{-1} T^{-2} / M^1 L^{-3} T^0]^{1/2} = [M^0 L^1 T^{-1}]$$

Thus, dimensions on both sides are the same, therefore the equation is correct.

Example 5. Using Principle of Homogeneity of dimensions, check the correctness of equation, $h = 2Td / rg \cos\theta$.

Sol. The given formula is, $h = 2Td / rg \cos\theta$.

Dimensions of term on L.H.S

$$\text{Height (h)} = [M^0 L^1 T^0]$$

Dimensions of terms on R.H.S

$$T = \text{surface tension} = [M^1 L^0 T^{-2}]$$

$$D = \text{density} = [M^1 L^{-3} T^0]$$

$$r = \text{radius} = [M^0 L^1 T^0]$$

$$g = \text{acc. due to gravity} = [M^0 L^1 T^{-2}]$$

$$\cos\theta = [M^0 L^0 T^0] = \text{no dimensions}$$

So,

$$\begin{aligned} \text{Dimensions of } 2Td/rg\cos\theta &= [M^1 L^0 T^{-2}] \times [M^1 L^{-3} T^0] / [M^0 L^1 T^0] \times [M^0 L^1 T^{-2}] \\ &= [M^2 L^{-5} T^0] \end{aligned}$$

Dimensions of terms on L.H.S are not equal to dimensions on R.H.S. Hence, formula is not correct.

Example 6. Check the accuracy of the following relations:

$$(i) \quad E = mgh + \frac{1}{2} mv^2; \quad (ii) \quad v^3 - u^2 = 2as^2.$$

Sol. (i) $E = mgh + \frac{1}{2} mv^2$

Here, dimensions of the term on L.H.S.

$$\text{Energy, } E = [M^1 L^2 T^{-2}]$$

Dimensions of the terms on R.H.S,

$$\text{Dimensions of the term, } mgh = [M] \times [LT^{-2}] \times [L] = [M^1 L^2 T^{-2}]$$

$$\text{Dimensions of the term, } \frac{1}{2} mv^2 = [M] \times [LT^{-1}]^2 = [M^1 L^2 T^{-2}]$$

Thus, dimensions of all the terms on both sides of the relation are the same, therefore, the relation is correct.

(ii) The given relation is,

$$v^3 - u^2 = 2as^2$$

Dimensions of the terms on L.H.S

$$v^3 = [M^0] \times [LT^{-1}]^3 = [M^0 L^3 T^{-3}]$$

$$u^2 = [M^0] \times [LT^{-1}]^2 = [M^0 L^2 T^{-2}]$$

Dimensions of the terms on R.H.S

$$2as^2 = [M^0] \times [LT^{-2}] \times [L]^2 = [M^0 L^3 T^{-2}]$$

Substituting the dimensions in the relations, $v^3 - u^2 = 2as^2$

We get, $[M^0 L^3 T^{-3}] - [M^0 L^2 T^{-2}] = [M^0 L^3 T^{-2}]$

The dimensions of all the terms on both sides are not same; therefore, the relation is not correct.

Example 7. The velocity of a particle is given in terms of time t by the equation

$$v = At + b/t + c$$

What are the dimensions of a, b and c?

Sol. Dimensional formula for L.H.S

$$V = [L^1 T^{-1}]$$

In the R.H.S dimensional formula of At

$$[T] = [L^1 T^{-1}]$$

$$A = [LT^{-1}] / [T^{-1}] = [L^1T^{-2}]$$

$t + c$ = time, c has dimensions of time and hence is added in t .

Dimensions of $t + c$ is $[T]$

Now, $b / t + c = v$

$$b = v(t + c) = [LT^{-1}] [T] = [L]$$

There dimensions of $a = [L^1T^{-2}]$, Dimensions of $b = [L]$ and that of $c = [T]$

Example 8. In the gas equation $(P + a/v^2)(v - b) = RT$, where T is the absolute temperature, P is pressure and v is volume of gas. What are dimensions of a and b ?

Sol. Like quantities are added or subtracted from each other i.e.,

$(P + a/v^2)$ has dimensions of pressure = $[ML^{-1}T^{-2}]$

Hence, a/v^2 will be dimensions of pressure = $[ML^{-1}T^{-2}]$

$$a = [ML^{-1}T^{-2}] [\text{volume}]^2 = [ML^{-1}T^{-2}] [L^3]^2$$

$$a = [ML^{-1}T^{-2}] [L^6] = [ML^5T^{-2}]$$

Dimensions of $a = [ML^5T^{-2}]$

$(v - b)$ have dimensions of volume i.e.,

b will have dimensions of volume i.e., $[L^3]$

$$\text{or } [M^0L^3T^0]$$

2. To convert a physical quantity from one system of units into another.

Physical quantity can be expressed as

$$Q = nu$$

Let n_1u_1 represent the numerical value and unit of a physical quantity in one system and n_2u_2 in the other system.

If for a physical quantity Q ; $M_1L_1T_1$ be the fundamental unit in one system and $M_2L_2T_2$ be fundamental unit of the other system and dimensions in mass, length and time in each system can be respectively a, b, c .

$$u_1 = [M_1^a L_1^b T_1^c]$$

and

$$u_2 = [M_2^a L_2^b T_2^c]$$

as we know

$$n_1u_1 = n_2u_2$$

$$n_2 = n_1u_1/u_2$$

$$n_2 = n_1 \frac{[M_1^a L_1^b T_1^c]}{[M_2^a L_2^b T_2^c]}$$

$$n_2 = n_1 \left[\left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c \right]$$

While applying the above relations the system of unit as first system in which numerical value of physical quantity is given and the other as second system

Thus knowing $[M_1L_1T_1]$, $[M_2L_2T_2]$ a, b, c and n_1 , we can calculate n_2 .

Example 9. Convert a force of 1 Newton to dyne.

Sol. To convert the force from MKS system to CGS system, we need the equation

$$Q = n_1 u_1 = n_2 u_2$$

$$\text{Thus } n_2 = \frac{n_1 u_1}{u_2}$$

Here $n_1 = 1$, $u_1 = 1\text{N}$, $u_2 = \text{dyne}$

$$n_2 = n_1 \frac{[M_1 L_1 T_1^{-2}]}{[M_2 L_2 T_2^{-2}]}$$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right) \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = n_1 \left(\frac{kg}{gm} \right) \left(\frac{m}{cm} \right) \left(\frac{s}{s} \right)^{-2}$$

$$n_2 = n_1 \left(\frac{1000gm}{gm} \right) \left(\frac{100cm}{cm} \right) \left(\frac{s}{s} \right)^{-2}$$

$$n_2 = 1(1000)(100)$$

$$n_2 = 10^5$$

Thus **1N = 10^5 dynes.**

Example 10. Convert work of 1 erg into Joule.

Sol: Here we need to convert work from CGS system to MKS system

Thus in the equation

$$n_2 = \frac{n_1 u_1}{u_2}$$

$$n_1 = 1$$

$u_1 = \text{erg}$ (CGS unit of work)

$u_2 = \text{joule}$ (SI unit of work)

$$n_2 = \frac{n_1 u_1}{u_2}$$

$$n_2 = n_1 \frac{M_1 L_1^2 T_1^{-2}}{M_2 L_2^2 T_2^{-2}}$$

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-2}$$

$$n_2 = n_1 \left(\frac{gm}{kg} \right) \left(\frac{cm}{m} \right)^2 \left(\frac{s}{s} \right)^{-2}$$

$$n_2 = n_1 \left(\frac{gm}{1000gm} \right) \left(\frac{cm}{100cm} \right)^2 \left(\frac{s}{s} \right)^{-2}$$

$$n_2 = 1(10^{-3})(10^{-2})^2 \quad n_2 = 10^{-7}$$

Thus, **1 erg = 10^{-7} Joule.**

Limitations of Dimensional Equation: The method of dimensions has the following limitations:

1. It does not help us to find the value of dimensionless constants involved in various physical relations. The values, of such constants have to be determined by some experiments or mathematical investigations.
2. This method fails to derive formula of a physical quantity which depends upon more than three factors. Because only three equations are obtained by comparing the powers of M, L and T.
3. It fails to derive relations of quantities involving exponential and trigonometric functions.
4. The method cannot be directly applied to derive relations which contain more than one terms on one side or both sides of the equation, such as $v = u + at$ or $s = ut + \frac{1}{2} at^2$ etc. However, such relations can be derived indirectly.
5. A dimensionally correct relation may not be true physical relation because the dimensional equality is not sufficient for the correctness of a given physical relation.

* * * * *

EXERCISES

Multiple Choice Questions

1. $[ML^{-1}T^{-2}]$ is the dimensional formula of
 - (A) Force
 - (B) Coefficient of friction
 - (C) Modulus of elasticity
 - (D) Energy.
2. 10^5 Fermi is equal to
 - (A) 1 meter
 - (B) 100 micron
 - (C) 1 Angstrom
 - (D) 1 mm
3. rad / sec is the unit of
 - (A) Angular displacement
 - (B) Angular velocity
 - (C) Angular acceleration
 - (D) Angular momentum
4. What is the unit for measuring the amplitude of a sound?

- (A) Decibel
- (B) Coulomb
- (C) Hum
- (D) Cycles

5. The displacement of particle moving along x-axis with respect to time is $x=at+bt^2-ct^3$.
The dimension of c is
- (A) LT^{-2}
 - (B) T^{-3}
 - (C) LT^{-3}
 - (D) T^{-3}

Short Answer Questions

1. Define Physics.
2. What do you mean by physical quantity?
3. Differentiate between fundamental and derived unit.
4. Write full form of the following system of unit
 - (i) CGS (ii) FPS (iii) MKS
5. Write definition of Dimensions.
6. What is the suitable unit for measuring distance between sun and earth?
7. Write the dimensional formula of the following physical quantity -
 - (i) Momentum (ii) Power (iii) Surface Tension (iv) Strain
8. What is the principle of Homogeneity of Dimensions?
9. Write the S.I & C.G.S units of the following physical quantities-
 - (a) Force (b) Work
10. What are the uses of dimensions?

Long Answer Questions

1. Check the correctness of the relation $\lambda = h /mv$; where λ is wavelength, h- Planck's constant, m is mass of the particle and v - velocity of the particle.
2. Explain different types of system of units.
3. Check the correctness of the following relation by using method of dimensions
 - (i) $v = u + at$
 - (ii) $F = mv / r^2$
 - (iii) $v^2 - u^2 = 2as$
4. What are the limitations of Dimensional analysis?
5. Convert an acceleration of 100 m/s^2 into km/hr^2 .

Answers to multiple choice questions:

- 1 (C) 2 (C) 3 (B) 4 (A) 5 (C)

ELECTROSTATICS

Learning Objectives : After studying this chapter, the student should be able to;

- Understand fundamental of charges at rest, properties of point charges;
- Explain conservation and quantization of charges;
- Relate the properties leading charge storage capacity of the electronic devices using static charges.

The branch of physics which deals with the study of charges at rest is called electrostatics.

9.1. ELECTRIC CHARGE

Electric Charge: it is the physical property of matter that causes it to experience force when placed in an electromagnetic field. There are two types of charges.

- (1) **Positive charge:** e.g. Proton, Alpha particle
- (2) **Negative charge:** e.g. Electron, etc.

Charge on electron is smallest unit of charge.

SI unit of charge is coulomb (C).

$$\text{Charge on Electron} = -1.6 \times 10^{-19} \text{ C}$$

$$\text{Charge on Proton} = +1.6 \times 10^{-19} \text{ C}$$

Like charges repel each other and unlike charges attract each other. i.e.

+ve	+ve	Repel
-ve	-ve	Repel
+ve	-ve	Attract
-ve	+ve	Attract

Conservation of Charge

Charge conservation is the principle that total electric charge in an isolated system never changes. It always remains constant. This also means that no net charge can be created or destroyed. When an atom is ionized, equal amounts of positive and negative charges are produced. Hence the algebraic sum of charges before and after remains the same.

Quantization of Charges

Charge quantization is the principle that the charge of any object is an **integer** multiple of the elementary charge (e). Thus, an object's charge can be exactly $0e$, or exactly $1e$, $-1e$, $2e$, etc.,

9.2. COULOMB LAW OF ELECTROSTATICS

It states that force of interaction between two point charges is

- (i) Directly proportional to magnitude of charges and

- (ii) Inversely proportional to the square of the distance between them.

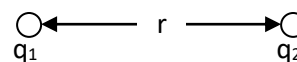
Let F is force between two charges q_1 and q_2 . Then

$$F \propto q_1 q_2$$

$$F \propto \frac{1}{r^2}$$

$$\Rightarrow F \propto \frac{q_1 q_2}{r^2} \quad \dots\dots\dots(1)$$

$$F = K \frac{q_1 q_2}{r^2} \quad \dots\dots\dots(2)$$



where K is constant of proportionality and its value is given as

$$K = \frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \text{ (in SI units system)}$$

Now from equation (2)

$$F = \frac{q_1 q_2}{4\pi \epsilon_0 r^2} \quad \dots\dots\dots(3)$$

Here ϵ_0 is electrical permittivity of vacuum. Its value is $8.854 \times 10^{-12} \text{ N}^{-1}\text{m}^{-2}\text{C}^2$.

Let $q_1 = q_2 = q$
 $r = 1 \text{ m}$

then from equation (3) $F = 9 \times 10^9 \text{ N}$

Thus *one coulomb is that much charge which produces a force of $9 \times 10^9 \text{ N}$ at a unit charge placed at a distance of 1 m.*

Smaller units of charge; milli Coulomb (mC) = 10^{-3} C .
 micro Coulomb (μC) = 10^{-6} C .

9.3. ELECTRIC FIELD

It is the area around the charge in which force of attraction or repulsion can be experienced by another charge.

Electric field intensity at point is defined as the force acting on a unit positive charge at that point.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

• A unit positive charge is also called as test charge

The value of q_0 should be very small. Its SI unit is N /C (Newton per Coulomb)

Electric Lines of Force:

An electric line of force is an imaginary continuous line or curve drawn in an electric field such that tangent to it at any point gives the direction of electric force at that point (Fig. 9.1).

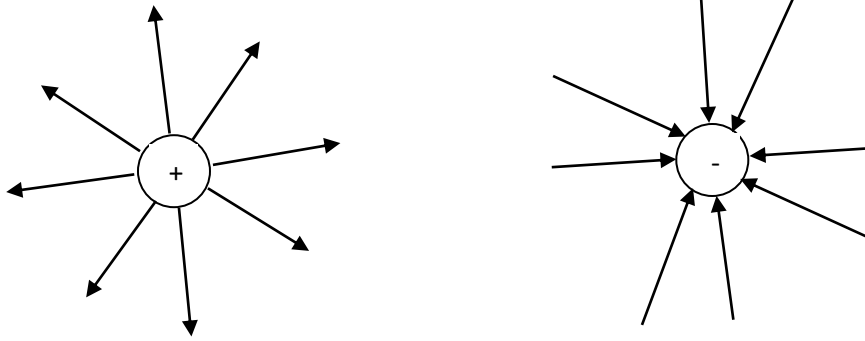


Figure 9.1

Properties of electric lines of force

- Lines of force originate from a positive charge and terminate to a negative charge.
 - The tangent to the line of force indicates the direction of the electric field and electric force.
 - Electric lines of force are always normal to the surface of charged body.
 - Electric lines of force contract longitudinally and expand laterally.
 - Two electric lines of force cannot intersect each other.
 - Two electric lines of force proceeding in the same direction repel each other.
 - Two electric lines of force proceeding in the opposite direction attract each other.
- There are no lines of force inside the conductor

9.4. ELECTRIC FLUX

It is the measure of distribution of electric field through a given surface. Electric flux is defined as total number of electric lines of force passing per unit area normal to the surface. It is denoted by ψ (psi).

Consider small elementary area $\vec{ds}$ on a closed surface S . Electric field $\vec{E}$ exit in the space. If θ is the angle between $\vec{E}$ and area vector $\vec{ds}$ as then

$$\psi = \oint \vec{E} \cdot \vec{ds} \text{ is called electric flux.}$$

GAUSS'S LAW

It states that net electric flux of an electric field over a closed surface is equal to the net charge enclosed by the surface divided by ϵ_0 i.e.

$$\psi = \oint \vec{E} \cdot \vec{ds}$$

$$\psi = \oint_S E dS \cos \theta = \frac{q}{\epsilon_0}$$

Proof: Consider a closed surface S having a charge q placed at a point O inside a closed surface as shown in Fig. 9.2. Take a point P on the surface and consider a small area dS around P .

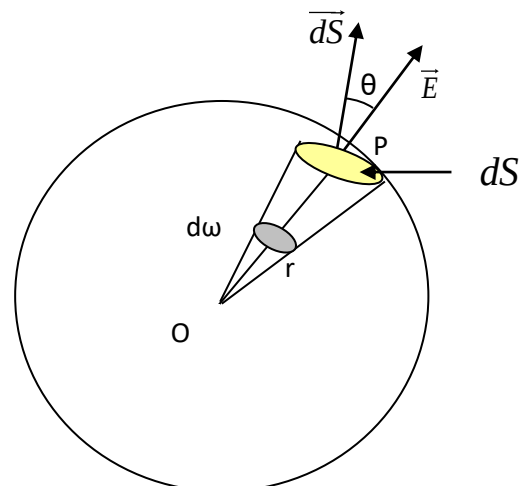


Figure 9.2

Then Electric field at P is

$$E = \frac{q}{4\pi \epsilon_0 r^2} \dots\dots\dots(1)$$

Now electric flux

$$\psi = \oint_S E dS \cos \theta$$

Putting value of E we get

$$\psi = \oint_S \frac{q}{4\pi \epsilon_0 r^2} dS \cos \theta$$

$$\psi = \frac{q}{4\pi \epsilon_0} \oint_S \frac{dS \cos \theta}{r^2}$$

$$\psi = \frac{q}{4\pi \epsilon_0} \oint_S d\omega$$

$$\psi = \frac{q}{4\pi \epsilon_0} . 4\pi$$

$$\psi = \frac{q}{\epsilon_0}$$

$$\text{Hence, } \psi = \oint_S E dS \cos \theta = \frac{q}{\epsilon_0}$$

$$\therefore \frac{dS \cos \theta}{r^2} = d\omega$$

$\therefore$ Total Solid angle $= 4\pi$

Applications of Gauss's Law:

Electric field due to a point charge:

Consider a point charge q . We want to find electric field at point P at a distance of r from it. Construct a spherical surface of radius r . This is called as Gaussian surface. Consider small area dS on the surface. Let θ is angle between $\vec{E}$ and Area vector as shown in Fig. 9.3.

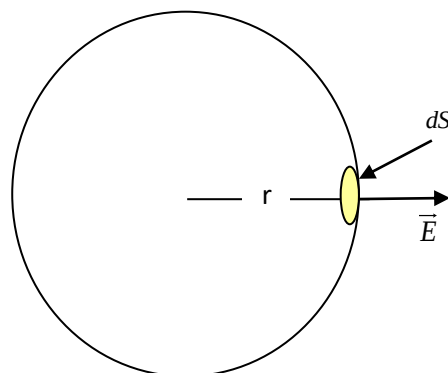


Figure 9.3

Now flux

$$\psi = \oint_S E dS \cos \theta = \frac{q}{\epsilon_0}$$

($\because \theta = 0$)

$$E \oint_S dS = \frac{q}{\epsilon_0}$$

$$\Rightarrow E . 4\pi r^2 = \frac{q}{\epsilon_0}$$

($\because$ Area of Sphere $= 4\pi r^2$)

$$\Rightarrow \boxed{E = \frac{q}{4\pi \epsilon_0 r^2}}$$

Thus the electric intensity decreases with increase in distance.

9.5. CAPACITOR

It is an electronic component that stores electric charge.

Capacitance

The ability of a system to store an electric charge.

As potential is proportional to charge

$$V \propto q$$

$$\text{or } q \propto V$$

$$q = CV$$

$$C = \frac{q}{V}$$

Unit of capacitance: Farad (F), microfarad

Grouping of Capacitors

Series Grouping:

A number of capacitors are said be connected in series if -ve plate of one capacitor is connected to the +ve plate of other capacitor and so on.

In this grouping, current is same on each capacitor.

Consider 3 capacitors of capacitances C_1, C_2, C_3 in series. Let V is total applied voltage.

$V_1, V_2, V_3 \rightarrow$ voltage drops across C_1, C_2, C_3 as shown in fig. 9.4.

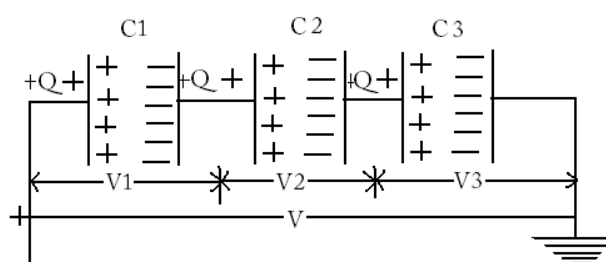


Figure 9.4

$$\text{Then } V = V_1 + V_2 + V_3 \text{ ----- (1)}$$

$$\text{Now } C = \frac{q}{V} \Rightarrow V = \frac{q}{C}$$

$$\text{So, } V_1 = \frac{q}{C_1}, V_2 = \frac{q}{C_2}, V_3 = \frac{q}{C_3}$$

Putting in Equation (1)

$$\frac{q}{C} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$$

$$\frac{q}{C} = q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

So the total capacitance decreases in series grouping.

The reciprocal of the equivalent capacitance of two capacitors connected in series is the sum of the reciprocals of the individual capacitances.

Parallel Grouping:

A number of capacitors are said to be connected in parallel if +ve plate of each capacitor is connected to the +ve terminal of battery and –ve plate of each capacitor is connected to the –ve terminal of battery.

In this grouping voltage across each capacitor is same.

Consider 3 capacitors of capacitances C_1, C_2, C_3 connected in parallel let V is applied voltage.

$q_1, q_2, q_3 \rightarrow$ charges on capacitors C_1, C_2, C_3 as shown in fig. 9.5

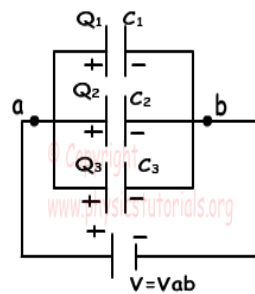


Figure 9.5

So

$$q = q_1 + q_2 + q_3 \text{ ----- (1)}$$

Now $C = \frac{q}{V}$ or $q = CV$

$$\therefore q_1 = C_1 V, \quad q_2 = C_2 V, \quad q_3 = C_3 V$$

Put in equation (1)

$$CV = C_1 V + C_2 V + C_3 V$$

$$CV = (C_1 + C_2 + C_3) V$$

$$C = C_1 + C_2 + C_3$$

So the total capacitance increases in parallel grouping.

The equivalent capacitance of capacitors connected in parallel is sum of the individual capacitance.

Solved Numericals

Example 1. Calculate the coulomb force between two protons separated by a distance of 1.6×10^{-15} m.

Solution: Given, 2 protons; Charge on Proton = 1.6×10^{-19} C

$$q_1 = q_2 = 1.6 \times 10^{-19} \text{ C}$$

$$\text{Distance, } r = 1.6 \times 10^{-15} \text{ m}$$

$$\text{Also } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\text{Now } F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$F = \frac{9 \times 10^9 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19}}{(1.6 \times 10^{-15})^2}$$

$$F = 90 \text{ N}$$

Example 2. Calculate the force between an alpha particle and a proton separated by distance of 5.12×10^{-15} m.

Solution : Given, q_1 = Charge on alpha particle = $2 \times 1.6 \times 10^{-19}$ C

$$q_2 = \text{Charge on proton} = 1.6 \times 10^{-19} \text{ C}$$

$$\text{distance, } r = 5.12 \times 10^{-15} \text{ m}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Now

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$F = \frac{9 \times 10^9 \times 3.2 \times 10^{-19} \times 1.6 \times 10^{-19}}{(5.12 \times 10^{-15})^2}$$

$$F = 17.58 \text{ N}$$

Example 3. Three Capacitors of capacitances $3\mu\text{F}$, $2\mu\text{F}$ and $4\mu\text{F}$ are connected with each calculate total capacitance (a) In Series grouping (b) Parallel grouping.

Solution: Given,

$$C_1 = 3\mu\text{F}, \quad C_2 = 2\mu\text{F} \quad \text{and} \quad C_3 = 9\mu\text{F}$$

In Series grouping

$$\frac{1}{C_{\text{tot}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{\text{tot}}} = \frac{1}{3} + \frac{1}{2} + \frac{1}{9}$$

$$= \frac{17}{18} \mu F$$

$$\therefore C_{\text{tot}} = \frac{18}{17} = 1.06 \mu F$$

In Parallel grouping

$$C_{\text{tot}} = C_1 + C_2 + C_3$$

$$C_{\text{tot}} = 3 + 2 + 9$$

$$C_{\text{tot}} = 14 \mu F$$

Example 4. Three capacitors 1F, 2F, and 3F are joined in series first and then in parallel. Calculate the ratio of equivalent capacitance in two cases.

Solution: Given,

$$C_1 = 1 \text{ F}, \quad C_2 = 2 \text{ F}, \quad C_3 = 3 \text{ F}$$

In series grouping

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_s} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3}$$

$$\frac{1}{C_s} = \frac{11}{6}$$

$$\therefore C = \frac{6}{11} \text{ F}$$

In Parallel grouping

$$C_p = C_1 + C_2 + C_3$$

$$C_p = 1 + 2 + 3$$

$$C_p = 6 \text{ F}$$

$$\therefore \text{Ratio} \quad \frac{C_p}{C_s} = \frac{6}{\frac{6}{11}}$$

$$\text{or} \quad \frac{C_p}{C_s} = 11$$

EXERCISES

Multiple Choice questions

- 1) Coulomb's law is only true for point charges whose sizes are
 - A. Medium
 - B. very large
 - C. very small
 - D. none of the above
- 2) As per Coulomb's law, force of attraction or repulsion between two point charges is directly proportional to
 - A. sum of the magnitude of charges
 - B. square of the distance between them
 - C. product of the magnitude of charges
 - D. cube of the distance
- 3) If F is force acting on test charge q_0 , electric field intensity E would be given by
 - A. $E = F - q_0$
 - B. $E = F/q_0$
 - C. $E = F + q_0$
 - D. $E = q_0/F$
- 4) In combination of capacitors in series, capacitors are connected
 - A. Parallel
 - B. side by side
 - C. up and down
 - D. none of the above
- 5) Coulomb's force between 2 point charges $10\mu\text{C}$ and $5\mu\text{C}$ placed at a distance of 150cm is
 - A. 0.2 N
 - B. 0.5 N
 - C. 2 N
 - D. 10 N
- 6) A device which stores charge is called
 - A. Resistor
 - B. Inductor
 - C. Capacitor
 - D. transistor
- 7) If medium between two charges is air, then value of constant k in SI units will be
 - A. $5 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
 - B. $7 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
 - C. $8 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
 - D. $9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
- 8) $1\text{ micro farad } (1\mu\text{F})$ is equal to
 - A. $1 \times 10^{-9}\text{ F}$
 - B. $1 \times 10^{-12}\text{ F}$
 - C. $1 \times 10^{-6}\text{ F}$
 - D. $1 \times 10^{-10}\text{ F}$
- 9) In equation $Q=CV$, C is constant of proportionality called capacitor's
 - A. Power
 - B. Capacitance
 - C. Heat

- D. Electric intensity
- 10) The force between two charges is 120 N. If the distances between the charges is doubled, the force will be
- A. 60 N
 - B. 30 N
 - C. 40 N
 - D. 15 N
- 11) SI unit of charge is
- A. Coulomb
 - B. Volt
 - C. Newton
 - D. Joule
- 12) The law governing force between electric charges is
- A. Gauss Law
 - B. Coulomb's law
 - C. Biot- Savart Law
 - D. Ampere's Law
- 13) Farad is the unit of
- A. Capacitance
 - B. Electric Potential
 - C. Force
 - D. Torque
- 14) Three capacitors of $2\mu\text{F}$ are joined in parallel; their resultant capacitance is
- A. $6\mu\text{F}$
 - B. 10nF
 - C. 6nF
 - D. $1.5\mu\text{F}$
- 15) Unit of Electric intensity is
- A. N/Coulomb
 - B. Joule
 - C. Newton
 - D. Coulomb
- 16) Two capacitors of 2 farad are joined in series their resultant is
- A. 1 F
 - B. 4 F
 - C. 3 F
 - D. 0.5 F

Short Answer Questions

1. Define Electric Field
2. What are Electric Lines of force ?
3. Define the term Capacitance.
4. What is Electric flux?
5. Define Capacitor
6. What do you mean by Electric Potential?
7. Define Electric Intensity.
8. State Coulomb's Law
9. Derive expression for electric intensity at a point due to point charge.

10. Explain properties of electric lines of force.
11. Calculate total capacitance in series combination.
12. Calculate total capacitance in parallel combination
13. Explain Gauss's law
14. Define Electric Charge and its types.

Long Answer Type Question

1. Calculate total capacitance when capacitors are connected in series and parallel grouping.
2. State and prove Gauss Law.
3. Using Gauss Theorem find electric field due to a point charge.
4. Derive Coulomb's Law.
5. Derive Coulomb's law from Gauss' law.
6. Use Gauss theorem to derive an expression for the electric field at a point due to a thin infinitely long straight line of charge of uniform charge density ?
7. Derive an expression for the electric field at a point due to uniformly charged spherical shell using Gauss' law.

Answers to Multiple Choice Questions

1)	C	2)	C	3)	B	4)	B
5)	A	6)	C	7)	D	8)	C
9)	B	10)	C	11)	A	12)	B
13)	A	14)	A	15)	A	16)	A